

Two-arms cGAS-dependence asymmetry across methods

IFN/ISG arm (top) is cGAS-dependent: significant, positive Interaction. HIF/glycolysis arm (bottom) rises in BOTH heat arms but is flat in the Interaction - no detectable cGAS-dependence at n=5. Sample mapping owner-confirmed (GSE329522 iTreg 2x2 genotype x temperature); n=5/group

|                                                                |                                                                                        | WT heat<br>(39 vs 37 °C) | cGAS-KO heat<br>(39 vs 37 °C) | Interaction | Heat main<br>(pooled) |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------|-------------------------------|-------------|-----------------------|
| IFN / ISG arm<br>(cGAS-dependent)                              | PROGENy JAK-STAT                                                                       | -6.5                     | -11.7                         | +16.8       | -9.3                  |
|                                                                | DE:limma-trend Irf7                                                                    | +0.9                     | -9.3                          | +7.2        | -6.0                  |
|                                                                | DE:limma-trend Oasl2                                                                   | -0.8                     | -10.6                         | +6.9        | -8.0                  |
|                                                                | DE:limma-trend Ifit1                                                                   | +4.2                     | -4.3                          | +6.0        | -0.1                  |
|                                                                | TF:CollecTRI Irf3                                                                      | -1.3                     | -3.2                          | +5.9        | -2.3                  |
|                                                                | TF:CollecTRI Stat2                                                                     | -0.8                     | -2.6                          | +5.8        | -1.7                  |
|                                                                | DE:limma-trend Mx1                                                                     | +5.7                     | -2.5                          | +5.8        | +2.2                  |
|                                                                | TF:CollecTRI Stat1                                                                     | +2.0                     | +0.3                          | +5.3        | +1.1                  |
|                                                                | DE:limma-trend Isg15                                                                   | +4.0                     | -2.4                          | +4.5        | +1.2                  |
|                                                                | TF:CollecTRI Irf1                                                                      | -0.6                     | -1.7                          | +3.4        | -1.2                  |
|                                                                | GSEA:Hallmark Interferon-alpha Response                                                | -1.9                     | -2.5                          | +3.1        | -2.2                  |
|                                                                | GSEA:Reactome Interferon-alpha beta Signaling                                          | -1.5                     | -1.8                          | +2.9        | -1.7                  |
|                                                                | GSEA:Hallmark Interferon-gamma Response                                                | -1.8                     | -2.2                          | +2.8        | -2.0                  |
|                                                                | GSEA:WikiPathways Type I Interferon Induction and Signaling During Sarscov2 Infection  | -1.4                     | -2.1                          | +2.7        | -1.8                  |
|                                                                | GSEA:GO_BP Antiviral Innate Immune Response                                            | +1.2                     | -1.4                          | +2.7        | -1.1                  |
|                                                                | GSEA:GO_BP Type I Interferon Mediated Signaling Pathway                                | -1.3                     | -1.8                          | +2.6        | -1.6                  |
|                                                                | GSEA:WikiPathways Hostpathogen Interaction of Human Coronaviruses Interferon Induction | -1.4                     | -1.8                          | +2.5        | -1.7                  |
|                                                                | GSEA:GO_BP Interferon Mediated Signaling Pathway                                       | -1.5                     | -1.8                          | +2.5        | -1.6                  |
|                                                                | GSEA:Reactome Interferon Signaling                                                     | -1.4                     | -1.7                          | +2.5        | -1.6                  |
|                                                                | GSEA:GO_BP Response to Type I Interferon                                               | -1.4                     | -1.8                          | +2.4        | -1.7                  |
| HIF / glycolysis arm<br>(no detectable cGAS-dependence at n=5) | GSEA:WikiPathways Angiogenesis Overview                                                | +0.9                     | -0.9                          | +1.8        | +0.9                  |
|                                                                | GSEA:Reactome Regulation of Pyruvate Metabolism                                        | -1.5                     | -1.9                          | +1.6        | -1.7                  |
|                                                                | GSEA:Reactome Signaling by VEGF                                                        | +1.5                     | +1.5                          | +1.4        | +1.5                  |
|                                                                | GSEA:KEGG VEGF Signaling Pathway                                                       | +1.6                     | +1.5                          | +1.4        | +1.6                  |
|                                                                | GSEA:GO_BP Regulation of Glucose Metabolic Process                                     | -1.4                     | -1.5                          | +1.3        | -1.4                  |
|                                                                | GSEA:WikiPathways Angiogenesis                                                         | +1.0                     | +0.9                          | +1.2        | +0.9                  |
|                                                                | GSEA:Reactome Pyruvate Metabolism                                                      | -1.2                     | -1.3                          | +1.2        | -1.3                  |
|                                                                | GSEA:WikiPathways Vegfavegfr2 Signaling                                                | +2.0                     | +1.9                          | +1.2        | +1.9                  |
|                                                                | GSEA:GO_BP Glucose Metabolic Process                                                   | +1.2                     | +1.1                          | +1.1        | +1.1                  |
|                                                                | GSEA:MitoXplorer Pyruvate Metabolism                                                   | -1.4                     | -1.4                          | +1.1        | -1.4                  |
|                                                                | GSEA:Reactome Vegfr2 Mediated Vascular Permeability                                    | +0.9                     | +1.0                          | +1.1        | +1.0                  |
|                                                                | GSEA:GO_BP Regulation of Cell Migration Involved in Sprouting Angiogenesis             | +1.6                     | +1.5                          | -0.9        | +1.5                  |
|                                                                | GSEA:WikiPathways Glycolysis and Gluconeogenesis                                       | +0.9                     | +1.0                          | -0.9        | +0.9                  |
|                                                                | GSEA:GO_BP Glycolytic Process Through Glucose 6 Phosphate                              | +1.1                     | +1.2                          | -1.1        | +1.2                  |
|                                                                | GSEA:Hallmark Glycolysis                                                               | +1.6                     | +1.6                          | -1.1        | +1.6                  |
|                                                                | GSEA:MitoXplorer Glycolysis                                                            | +1.3                     | +1.4                          | -1.1        | +1.3                  |
|                                                                | GSEA:KEGG Pyruvate Metabolism                                                          | -1.3                     | -1.2                          | -1.1        | -1.2                  |
|                                                                | GSEA:KEGG Glycolysis Gluconeogenesis                                                   | +1.0                     | +1.1                          | -1.3        | +1.0                  |
|                                                                | GSEA:Hallmark Hypoxia                                                                  | +1.9                     | +2.0                          | -1.3        | +2.0                  |
|                                                                | PROGENy Hypoxia                                                                        | +4.0                     | +4.6                          | -2.2        | +4.4                  |